

Roll No.

TCS-710

B. TECH. (ML, DS, CC + IS, SE)

(SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

STORAGE NETWORKS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) A hospital uses an application that stores patient X-ray data in the form of large binary objects in an oracle database. The application is hosted on UNIX server, and the hospital staff accesses the X-ray records through a Gigabit Ethernet backbone. Storage array provides storages to the UNIX server, which has 6 terabytes of usable capacity. (CO1)

Describe how the value of this patient data might change over time.

OR

- (b) Define server centric IT architecture and storage centric IT architecture with advantages and limitations. (CO1)
2. (a) What are the performance characteristics of hard disk drive ? Explain. (CO2)

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(2)

OR

(b) Discuss the need of Data Center Infrastructure. Illustrate core elements of Data Center (Refer Q. 1 a). (CO2)

3. (a) List the factors contributed to the growth of digital data. (CO2)

OR

(b) Discuss the challenges in information management. (CO2)

4. (a) Illustrate the types of RAID implementation with their merits and demerits. (CO3)

OR

(b) Illustrate the different techniques to determine the data availability and performance characteristics of array. (CO3)

5. (a) Explain the importance of information Lifecycle Management. List the activities in developing the ILM strategy. (CO3)

OR

(b) Characterize and explain the performance benefits of the following nested RAID : (CO3)

